

OC_02: Order Basic Payment Bundle - IBAN Reservation

Public Digital PSO Solution Architect

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HL Journey	I Order Basic Payment Bundle

1 Introduction

1.1 Purpose and Scope

This document extends **OC_01 - Order Basic Payment Bundle - Order Capture & Product Inventory**, providing customers with the capability to apply for a new Basic Payment Bundle within the banking system.

It defines the functionalities, business activities, and **DigitalONE** operations required to support additional steps in the **Apply New Payment Bundle** process, allowing customers to reserved IBAN.

Line Of Business

Payment Account

Customer Segment

B2C (Business-to-Consumer) Banking Customer

Channels

Internet Banking (Self-Service)

1.2 Business Needs

As an existing customer, I want to apply for a new **Basic Payment Account Bundle**, reserve an **IBAN**.

1.3 Application Involved

Application	Description	Provider	Impacted
Catalog ONE	Amdocs Product Catalog (Catalog ONE) enables the management of offers, products, and accessories. It defines product characteristics, rules, pricing policies, and eligibility conditions.	Amdocs	Yes
Digital ONE	Amdocs Digital ONE provides a set of value-based, self-contained business functions which support diverse commerce and care workflows. New operations will be introduced to facilitate IBAN reservation and agreement generation .	Amdocs	Yes

Application	Description	Provider	Impacted
ABN AMRO Systems	ABN AMRO systems responsible for IBAN Reservation • CRC - Centrale Rekening Courant- Central administration for banking account	ABN AMRO	No

1.4 Related Document

Document ID	Type	Document Name
OC_01	Word	Order Basic Payment Bundle - Order Capture & Product Inventory

1.5 Assumptions

ID	Item	Description
01	Business Flow	The business flows and diagrams in this document reflect best practices and may be subject to change in future epics. However, backend components shown in these diagrams are within the committed scope. The document outlines the logical steps and integration points with Amdocs Microservices and ABN AMRO systems. These flows do not correspond to a specific UI application. Any screenshots included are for illustrative and reference purposes only.
03	New D1 Microservices	The operation name, main inputs, and outputs of the new D1 microservices described in this document may change as the integration progresses and the design is finalized.
04	Reserve IBAN	For the Reserve Resource operation, DigitalONE supports CRC Integration for IBAN reservation. However, the D1 microservice is designed to easily support additional resource types & Integrations beyond IBAN & CRC , in accordance with ABN AMRO's future requirements.

ID	Item	Description
05	IBAN Life Cycle	IBAN resource management and Life Cycle is managed by CRC. For successful order process, the resource returned assumed to be available for such order reservation.
06	Release IBAN	If, for any reason, the IBAN reservation succeeds but the update to the product order fails, the operation will return a failure. Rollback for releasing IBAN (if required) will be supported in a future phase.
07	IBAN Operation	Reserve IBAN can be invoked at any stage during order capture , as long as it happens after the order is created and before agreement generation . Consider the business Req discussed, it is recommended to invoke the Reserve IBAN operation right before agreement generation , rather than at the beginning
07	C1 Manual Selection	The C1 Manual Selection indicator for the IBAN Resource is intended for future use .
08	IBAN Value	The IBAN value which represent Client Selection in the Reserve Resource Operation is intended for future use .

1.6 Terms & Definitions

Term	Definition
C1 / CatalogONE	Amdocs CatalogONE is an online product catalog based on TM Forum standards. It enables the management of commercial offerings and product definitions.
D1 / DigitalONE	Amdocs DigitalONE consists of a set of functional capabilities used to handle the ordering flow, including order capture and fulfillment.
OC	Amdocs D1 Order Capture (OC) solution consists of a set of functional capabilities used to handle the negotiation of the order before submission
AOM	The Amdocs Order Management (AOM) solution consists of a set of functional capabilities used to handle the order after submission and for Amdocs Order Fulfillment.

2 High Level Solution

2.1 Overview

This document describes the additional steps in the **Apply New Payment Bundle** process, enabling customers to reserved **IBAN**, access the **bundle agreement**, and submit the order for fulfillment.

2.1.1 CatalogONE

New IBAN resource will be added to **CatalogONE** for the Payment Account product to enable **IBAN reservation**, including validations such as ensuring the IBAN is reserved before submission.

2.1.2 Order Capture

The order capture process is responsible for discovering and gathering the necessary information to create and submit an order, as outlined in **OC_01 - Order Basic Payment Bundle - Order Capture & Product Inventory**.

The **Amdocs Commerce** component will introduce new operations to support the additional activities in the end-to-end (E2E) order capture sales process.

The following new activities, highlighted in **green**, will be incorporated into the order flow to support IBAN Reservation:

**The order steps listed below are provided for clarity and reference purposes only. Those will be confirmed in Epic Exp_01-*

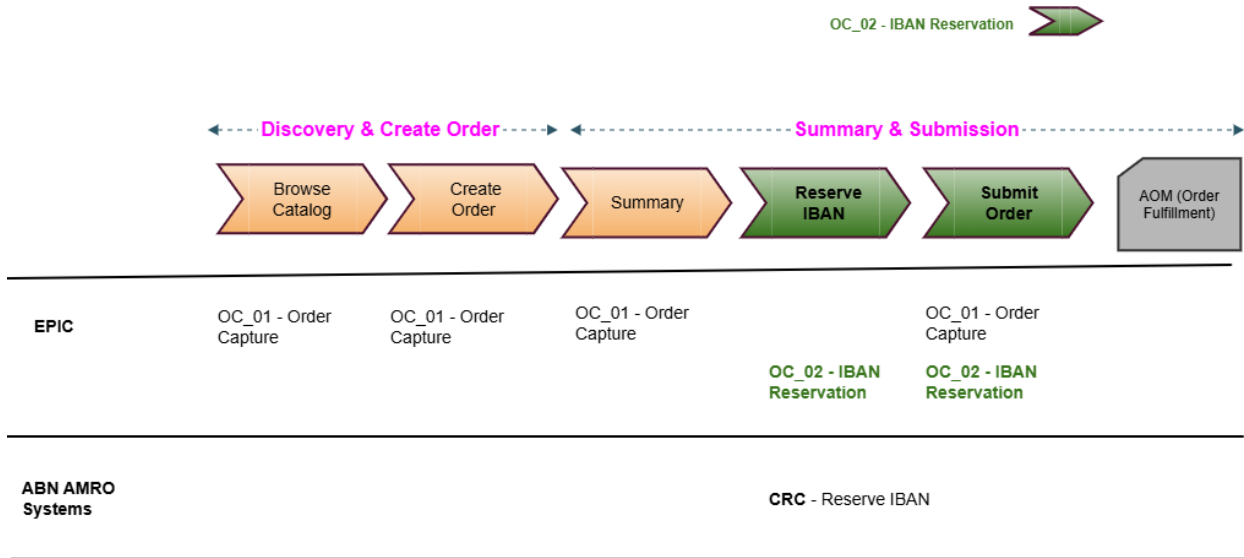
- **Browse Catalog** – Navigate and find a suitable bundle offering.
- **Order Creation** – Create an order with the selected bundle and its associated products.
- **Summary** – Review order details, including the bundle and its associated products.
- **Reserve IBAN** – Reserve the **IBAN** through **CRC Integration** and associate it with the corresponding **payment account** in the order.
- **Order Submission** – Submit the order for fulfillment.

2.2 Business Flows

2.2.1 Flow Diagram

The **Activity** steps in the diagram are provided for clarity and reference purposes only.

Order Capture - Order Basic Payment Bundle



2.2.2 Flow Description

The Order Capture process solution consists of a set of functional steps that are used to capture order details and submit the order for fulfillment.

The following new steps, highlighted in **green**, will be incorporated into the order flow:

Step	Description	Commerce Sub Domain	EPIC
Browse Catalog	Browse Bundle enable customer for viewing the catalog data and finding a suitable bundle.	Product Offering Discovery	OC_01 - Order Capture & Product Inventory
Create Order	Create an order that includes the selected Bundle Product Offering and its associated products. Before creating the order, validation will be performed.	Product Ordering Management	OC_01 - Order Capture & Product Inventory
Summary	The summary capability offers a comprehensive view of all order items, including their prices (if applicable), before the final order is submitted	Product Ordering Management	OC_01 Order Capture & Product Inventory

Step	Description	Commerce Sub Domain	EPIC
Reserve IBAN	Reserve the IBAN for the Payment Account	AABNL Product Ordering Management	OC_02 - IBAN Reservation
Submit Order	Submit the order and send the order for fulfillment. Ensure the IBAN was reserved, before submission	Product Ordering Management	OC_01 - Order Capture & Product Inventory OC_02 - IBAN Reservation

2.3 Business Use Cases

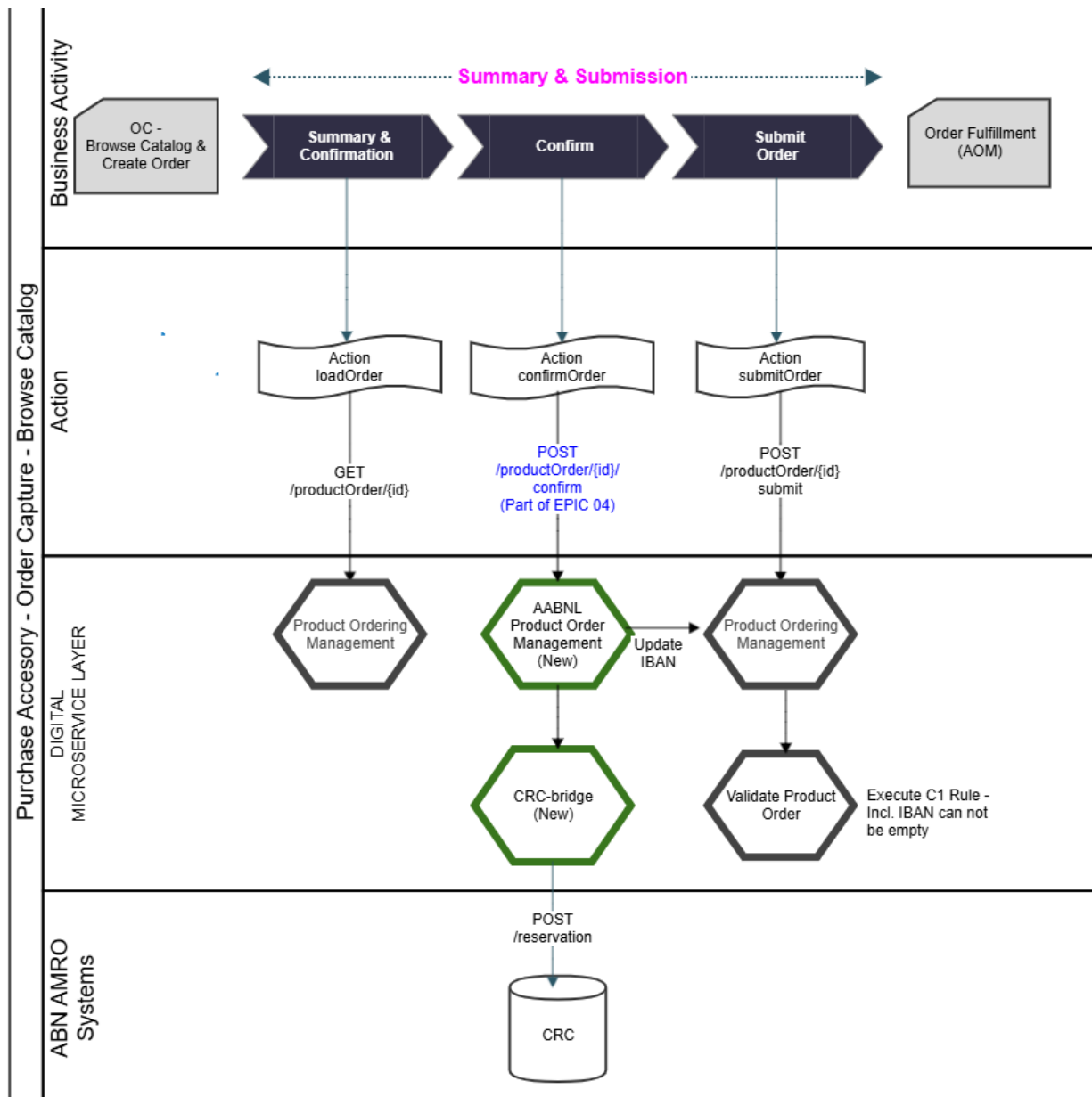
Use Case ID	Description	Expected Outcome
#01	The customer orders a Basic Payment Account Bundle, creates an order with the bundle, reserve the IBAN, and submits the order for fulfillment. If IBAN was not reserved before order submission, error will be shown to user and order will not be submitted.	The customer journey should allow them to: • Browse Catalog – Navigate and view available bundle offerings. • Order Creation – Select the Basic Payment Account Bundle and create an order. • View Summary – Review the order details including • Reserve IBAN - Reserve the IBAN for the Payment Account • Order Submission – Submit the order for fulfillment.

3 D1 Order Capture - Capabilities

3.1 Flow Diagram

3.1.1 Diagram

The **Business Activity** steps in the diagram are provided for clarity and reference purposes only.



3.1.2 Action Integration Point

Action	Commerce Sub Domain	REST API	Description
loadOrder	Product Order Management	GET /productOrder/{id}	Retrieve the order details

Action	Commerce Sub Domain	REST API	Description
confirmOrder	AABNL Product Order Management	POST / productOrder/{id}/ confirm (part of EPIC 04)	Reserve the IBAN through CRC Integration and associate it with the corresponding payment account in the order.
submitOrder	Product Order Management	POST / productOrder/{id}/ submitOrder	Submits a product order entity to Amdocs Order Management for its fulfillment, after performing the preconfigured required validations. The state of the product order entity is set to <i>Submitted</i> .

3.2 Solution Details

The following is the new operation to support the **Reserve IBAN** business activity.

3.2.1 Reserve IBAN

3.2.1.1 Overview

Operation	POST /aabnlBridge/crc/iban/v1/reserve
D1 Microservice	CRC Bridge AABNL - Microservice
Overview Details	Reserve a resource by type and associate it with the corresponding product in the order. The supported resource type is IBAN, and in the future, additional resource types may be introduced based on system requirements.
Main Input	

3.2.1.2 Description

The Resources layer of the microservice represents the exposed service endpoints in terms of REST services, and the contract of the component. The microservice includes the following resources:

- POST /aabnlBridge/crc/iban/v1/reserve

A new operation will be introduced to reserve a specific resource type

Supported Resource Type: IBAN with CRC Integration

- Defined in C1 as a Resource Characteristic with predefined properties:
 - Resource Type = IBAN
 - Provider = CRC
 - Account Type
- Part of the Payment Account product structure, designated as the Service ID of the product.

CRC Reservation Process

- For IBAN Reservation, **DigitalONE** will integrate with the **CRC reservation API**
- Once successfully reserved, the **IBAN resource value** will be persisted in the **Payment Account** product (part of EPIC 04 -confirm operation)
- If the required **resource already exists** on the **Payment Account**, it will skip the reservation (part of EPIC 04 -confirm operation)
- Reservation is not allowed if the product order is not in draft state (configuration in D1 - list of allowed statuses).

3.2.1.3 Integration Point

Functionality	Application	Operation	Description	Provider	Consumer	Interface Type	Mode	Integration Direction
Reserve	CRC	POST / reservation	Reserve a resource	ABN AMRO	Amdocs	REST	Sync	Southbound

3.2.1.4 Acceptance Criteria

Sunny Day Scenario

Test Name	GIVEN	WHEN	THEN
Reserving IBAN	Given input is provided in the request	When the system processes the reservation request	Then it should automatically reserve the unreserved resources (one resource)
IBAN Already Reserved for Payment Account (part of EPIC 04)	Given the Payment Account already contains a reserved IBAN	When a reservation request is made	Then the system should skip the reservation process for that BAN
Reserving an IBAN when the product order is in draft state (part of EPIC 04)	Given the product order is in draft state	When a request is made to reserve an IBAN	Then the system should proceed with the reservation process

Rainy Day Scenario

Test Name	GIVEN	WHEN	THEN
IBAN Reservation Fails Due to CRC Integration Issue	Given the customer has requested a new payment account AND the D1 new operation (POST .. \reserveResource is invoked with the product order ID as input	When the IBAN has not been previously reserved for this account And the system attempts to reserve the IBAN via CRC Integration And the CRC Integration API fails to process the IBAN reservation due to an internal error (e.g., connectivity issue, system downtime, or invalid API response)	Then an error is returned.

Test Name	GIVEN	WHEN	THEN
IBAN Reservation Fails Due to, order is not in draft state (part of EPIC 04)	Given the customer has requested a new payment account AND the D1 new operation (POST .. \reserveResource is invoked with the product order ID as input	When product order that is not in the draft state	Then the system should reject the request and return an error indicating that reservations are only allowed in the draft state

3.2.1.5 Screenshot (For Illustration only)

N/A

4 Product Catalog

4.1 IBAN Resource

IBAN Resource will be defined as part of the Resource Management policy within the **Payment Account** Product Spec Type in CatalogONE.

Characteristics: IBAN

Property	Value
Base Type	Logical
Resource Type	IBAN
Provider	CRC
Account Type	"CURRENT_ACCOUNTS"
Service ID	True
Manual Selection	False (For Future Use)

New Rule

IBAN will be defined as an optional characteristic.

A new **product rule** will be activated for all **Payment Accounts**, generating an error message if the IBAN field is empty upon validate & submit order (for order in draft status).

5 ABN AMRO Systems

5.1 Overview

Following are the ABN AMRO systems used by Amdocs Ordering Systems for IBAN Reservation and agreement generation

Application	Interface Description	Provider OAR ID	Provider	Consumer	Integrations Direction
CRC Centrale Reeking Courant	Central administration for current accounts.	AAB.SYS.1175	ABN AMRO	Amdocs Ordering	Southbound

5.2 Operations

5.2.1 POST /reservation

5.2.1.1 Overview

Operation	POST /reservation
Provider	CRC
Interface Type / Mode	REST / Sync
Description	This operation is used to reserve an IBAN for a customer.
Additional Information	

5.2.1.2 Main Input

Parameter name	Mandatory/Optional	Description	Example	Additional Information
customerId	Optional	Unique identifier of a customer for which new payment account belong to	76869763	
accountType	Mandatory	The type of the account to be created - CURRENT_ACCOUNTS - FISCAL_SAVINGS_ACCOUNTS - TERM_DEPOSIT_ACCOUNTS	CURRENT_ACCOUNTS	Value in C1 (Param List for CRC Resource)

5.2.1.3 Main Output

Parameter name	Description	Example	Additional Information
IBAN	International Bank Account Number	NL81ABNA0456279342	

6 Operational Aspects

6.1 Non-Functional

6.2 Authorization & Security

Will be handled in a separate EPIC

6.3 Multi-lingual

Will be handled in a separate EPIC

6.4 Error Handling

The D1 Reserve operation may return partial success in the case of multiple resource allocations.

- Rollback for releasing IBAN will be supported in a future phase.

6.5 Performance

N/A

6.6 Migration Impact

Not Applicable.

7 Appendix

7.1 Open Items / Action Items

I D #	Type	Description	Comment / Discussion Point	Date Created	St at us	Owner
1						

7.2 Release Notes

Version	Author	Date	Comments
Ver 0.01	Dana Roffe	12-March-2025	First Release
Ver 1	@Dana Roffe	23-March-2025	Final Version - Ver 1
V 1	@Dana Roffe	30-March-2025	update the name of the operation to: POST /produtOrder/{id}/resource/Reserve

8 Feature IBAN Reservation

8.1 Product Catalog

8.1.1 IBAN Resource

IBAN Resource will be defined for the **Payment Account** product in CatalogONE

Characteristics: IBAN

Property	Value
Resource Type	IBAN
System	CRC
Account Type	CURRENT_ACCOUNT
Service ID	True

New Rule

IBAN is mandatory upon submission.

8.2 D1 New Operations

8.2.1 AABNL Product Ordering Management MS

8.2.1.1 POST /produtOrder/{id}/reserveResource

8.2.1.1.1 Description

The **Reserve Resource** capability will be used to **reserve a resource by type** and associate it with the corresponding **product** in the order.

The **default resource type** is **IBAN**, and in the future, additional resource types may be introduced based on system requirements.

8.2.1.1.2 Technical Description

The capability will perform the following:

- Search within the product structure for the **IBAN resource**
 - Characteristic with **resource type = IBAN** (property definition)
- Retrieve the necessary integration property values (**System & Account Type**)
- Invoke the integration accordingly

For **CRC System**, **DigitalONE** will invoke the **CRC reservation API** via **CRC Central Banking Account MS** to reserve the IBAN, passing the following details:

- **Customer ID** associated with the order
- **Account Type** value as specified by the property in **CatalogONE** (e.g. Account Type=**CURRENT_ACCOUNT**)

Once the IBAN is successfully reserved, the IBAN value will be persisted on the **Payment Account product**.

If the resource required, already exists on the **Payment Account**, it will be returned without invoking CRC reservation operation.

If the product order is not in draft state, reservation is not allowed.

CRC Central Banking Account MS - will include 2 operations: reserve (used by OC) and activate (used by OH)

support partial success, error message - list of order item id which success/fails

8.2.1.1.3 Main Input

Attribute	Mandatory/ Optional	Description	Additional Information
Product Order ID	Mandatory		
Resource Type	Optional		Default = "IBAN"

8.2.1.1.4 Main Output - maybe not needed

Attribute	Description	Additional Information
Resource List []		

Attribute	Description	Additional Information
Order Item	The Order item ID	
Resource Number	The Resource Number (e.g. IBAN)	e.g. NL81ABNA0456279342
Type	The Type of the Resouce	e.g. IBAN